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Fully oxygen-tolerant reversible complexation mediated polymerization without external additives

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ABSTRACT

Reversible complexation mediated polymerization (RCMP) is one of the most robust reversible deactivation radical polymerization (RDRP) systems. However, oxygen sensitivity hinders its practical applications in conventional RCMP. In this study, the fully oxygen-tolerant RCMP is successfully carried out under visible light irradiation using alkyl iodides as initiators and tertiary amines (TAs) as catalysts. This open-to-air RCMP requires neither additives nor prior deoxygenation. Furthermore, the polymerization exhibited first-order kinetics, and molecular weight increased linearly with monomer conversion, demonstrating "living" characteristics under open condition. We believe this minimalist method can offer new opportunities for the application of RCMP in an aerobic environment.

1. Introduction

Reversible deactivation radical polymerization (RDRP), including atom transfer radical polymerization (ATRP), [1–4] reversible addition-fragmentation chain transfer (RAFT) polymerization, [5,6] nitroxide mediated radical polymerization (NMP) [7,8] and reversible complexation mediated polymerization (RCMP), [9,10] can be successfully employed to prepare a variety of polymeric materials with controllable molecular weight, narrow molecular weight distribution and various topological structures. However, traditional RDRPs in general needs to be carried out under an oxygen-free atmosphere, as oxygen (O_2) is a radical scavenger that could rapidly quench propagating radicals, leading to loss of control over molecular weights and molecular weight distributions [11–14]. To overcome this limitation, much effort has been devoted to developing fully oxygen-tolerant ATRP and RAFT polymerization without prior deoxygenation. For example, various redox-active catalysts were added to remove oxygen in RDRPs [15–23]. On the other hand, enzyme such as glucose oxidase (GO_x) or pyranose oxidase (P_2O_x) were employed to convert O_2 to inert species, [24–26] facilitating polymerization to be carried out without physical degassing. Besides, trialkylborane employed as co-initiators could be oxidized by oxygen, generating carbon-centered radicals to initiate polymerization in open air [27,28]. In our previous work, we have successfully developed fully oxygen-tolerant RDRPs employing magnetic $Zn_{0.64}Fe_{2.36}O_4$ nanoparticles [29] or copper(II) thioxanthone carboxylate ($Cu(TX)_2$) [30] as oxygen scavengers.

Compared with ATRP and RAFT polymerization, reversible complexation mediated polymerization (RCMP) employing alkyl iodides (R-I) as initiators and tertiary amines (TAs) and salts as catalysts, has gained more and more attention due to its distinct features, such as mild conditions, wide monomer versatility and the absence of metal catalysts or odorous compounds [31–33]. Although the unstable C-I bond of alkyl iodides brought challenges due to the relatively low bond dissociation energy (BDE), in situ bromine-iodine transformation strategies were successfully developed via halogen exchange reaction of alkyl bromide (R-Br) with sodium iodide (NaI), leading to excellent control of polymerization [34–37]. However, it should be noted that deoxygenation is required before polymerization in most cases, there are few studies on fully oxygen-tolerant RCMP. In 2022, Goto and co-workers firstly reported use of *N*-hydroxyphthalimide (NHPI) as an oxidation catalyst for converting an aldehyde to a carboxylic acid, where O_2 can be consumed in situ during polymerization, enabling RCMP to proceed in open air.

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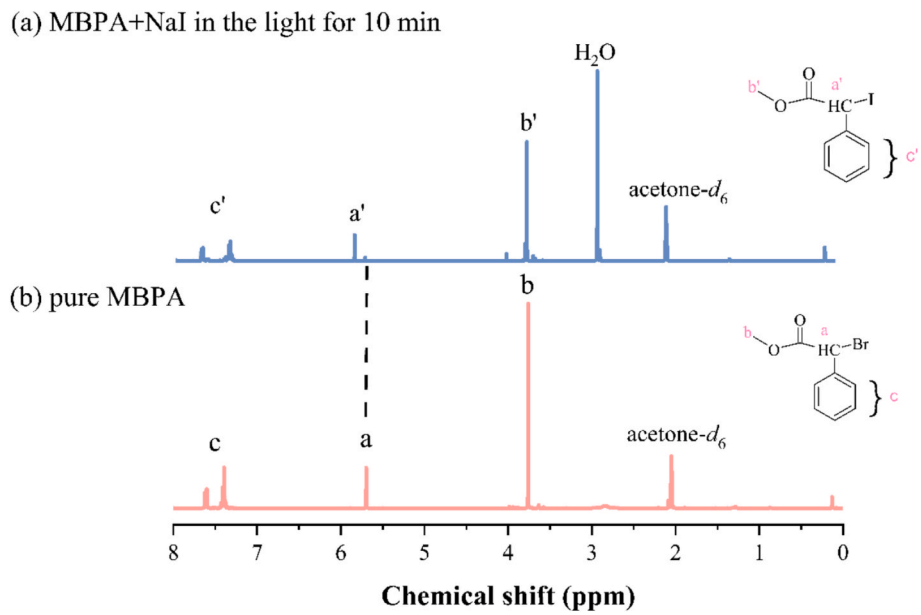

Fig. 1. ^{1}H NMR spectra of acetone-d_6 solutions of (a) MBPA (0.02 mmol) and NaI (0.08 mmol) at room temperature under visible light for 10 min, and (b) pure MBPA.

Notably, apart from NHPI, the base should be added to generate a carboxylate anion, which is a key agent for regulating this complex polymerization process [38].

It is well-known that tertiary amines (TAs) are widely employed as ligands or catalysts in ATRP and RAFT polymerization due to their unique electronic structure [39,40]. In the case of oxygen-tolerant RDRPs, TAs served as reducing agent or cocatalyst could consume oxygen by photoinduced electron transfer [41]. Qiao *et al.* found that trithiocarbonate RAFT agents irradiated with visible light could significantly reduce the amount of dissolved oxygen in the presence of TA [42]. Cheng and co-workers reported that *N,N,N′,N″,N″*-pentamethyldiethylenetriamine (PMDETA) could be used as an efficient catalyst for heat-induced RCMP of methyl methacrylate (MMA) in the presence of residual oxygen [43]. Our group has reported the oxygen-tolerant RAFT polymerization system using 2-(diethylamino)ethyl methacrylate (DMAEMA) as a monomer and an electron donor [44]. In these systems, tertiary amine plays an important role in oxygen removal. Therefore, we hypothesized that it should be possible to perform fully oxygen-tolerant reversible complexation mediated polymerization without additives or catalysts. Herein, the fully oxygen-tolerant RCMP was successfully developed with an assistant of TA. Notably, photoinduced tertiary amine reduction strategy is extremely simple, requiring no additives other than monomer, alkyl iodide and TA.

2. Experimental section

2.1. Materials

All monomers were purchased from Shanghai Aladdin Biochemical Technology Co, LTD, and were passed through a basic alumina column to remove the inhibitor. *N,N,N',N',N''*- pentamethyldiethylenetriamine (PMDETA), tris[2- (dimethylamino)ethyl]amine (Me_6TREN) and *N,N, N',N'*-Tetramethylethylenediamine (TMEDA) were purchased from Meryer and used as received. Dimethyl sulfoxide (DMSO), sodium iodide (NaI), methyl α-bromophenylacetate (MBPA) were purchased from Aladdin. Tetrabutylammonium iodide (TBAI) was purchased from Zancheng (Tianjin) Technology Co. 2-iodo-2-methylpropanenitrile (CP-I) was purchased from Adamas. Petroleum ether, tetrahydrofuran (THF), *N, N*-dimethylformamide (DMF), *N,N*-dimethylacetamide (DMAC) and methanol were purchased from Jiangsu Qiangsheng Functional Chemical Co, LTD.

2.2. Characterization

The number-average molecular weight (M_n) and molecular weight distribution (M_w/M_n) of the obtained polymers were determined on an Agilent PL-GPC 50 gel permeation chromatography (GPC) equipped with a refractive index detector, using a 5 μm Guard, 5 μm MIXED-D column with PMMA standard samples and 0.05 mol L^{-1} lithium bromide solution in DMF used as the eluent operated at 50 °C. The ^{1}H NMR of the polymers were recorded on an INOVA 300 MHz nuclear magnetic resonance instrument using acetone-d_6 or DMSO-d_6 as the solvent and tetramethylsilane (TMS) as an internal standard. Matrix assisted laser desorption ionization time of flight mass spectrometry was performed by Bruker Ultraflex TOF mass spectrometer using DCTB as matrix and sodium trifluoroacetate as auxiliary salt. UV–vis spectra were recorded on a UV-3600 made by SHIMADZU corporation, whose wavelengths range from 200 nm to 800 nm.

2.3. Procedure of bromine-iodine transformation

A typical procedure for bromine-iodine transformation: MBPA (5.4 mg, 0.023 mmol) and acetone-d_6 in an NMR tube was measured to acquire ^{1}H NMR spectra. Then, the mixture was transferred to a Schlenk tube with subsequent addition of NaI (14.0 mg, 0.093 mmol), stirred and irradiated with the light for 10 min, then transferred to the NMR tube again to obtain the ^{1}H NMR spectra.

2.4. Fully oxygen-tolerant RCMP

A typical procedure for the fully oxygen-tolerant RCMP was as follows. MMA (468 mg, 4.67 mmol), MBPA (5.4 mg, 0.023 mmol), PMDETA (4.1 mg, 0.023 mmol), NaI (14.0 mg, 0.093 mmol) and 0.5 mL DMAC were placed in a 2 mL ampoule bottle. The solution was not sealed in an open ampoule bottle. Open-to-air polymerizations were carried out under direct light illumination at 25 °C. The visible light source used in the experiment was a household lamp (12 W, 4 mW/cm^2). After the amount of reaction time, the samples were diluted with 2 ml THF and precipitated with 250 mL methanol. The polymer was obtained by filtration and dried to keep constant weight under vacuum.

Table 1
Fully Oxygen-Tolerant RCMPs with Different Catalysts.

Entry	Polymerization condition[a]	Time (h)	Conv (%)	M_n ($M_{n,th}$)	M_w/M_n
1	$[MMA]_0$:$[MBPA]_0$:$[NaI]_0$:$[TMEDA]_0$ = 200:1:4:1	24	7.2	19300 (1400)	1.34
2	$[MMA]_0$:$[MBPA]_0$:$[NaI]_0$:$[Me_6TREN]_0$ = 200:1:4:1	21	71.2	25500 (14200)	1.38
3	$[MMA]_0$:$[MBPA]_0$:$[NaI]_0$:$[PMDETA]_0$ = 200:1:4:1	21	74.3	26300 (15100)	1.15
4	$[MMA]_0$:$[MBPA]_0$:$[TBAI]_0$:$[Me_6TREN]_0$ = 200:1:4:1	12	58.6	29000 (12000)	1.31
5	$[MMA]_0$:$[MBPA]_0$:$[TBAI]_0$:$[PMDETA]_0$ = 200:1:4:1	12	49.1	30400 (10100)	1.28
6	$[MMA]_0$:$[CP\text{-}I]_0$:$[TMEDA]_0$ = 200:2:1	24	12.5	19700 (1300)	1.26
7	$[MMA]_0$:$[CP\text{-}I]_0$:$[Me_6TREN]_0$ = 200:2:1	21	62.7	18400 (6300)	1.22
8	$[MMA]_0$:$[CP\text{-}I]_0$:$[PMDETA]_0$ = 200:2:1	5	74.9	9200 (7700)	1.32

All polymerizations were performed under visible light irradiation (12 W, 4 mW/cm^2) at 25 °C and in DMAC. The volume ratio of DMAC to MMA was maintained at 1:1 for all entries.

2.5. UV–vis spectroscopy.

A typical procedure for a UV–vis study was as follows. CP-I (9.1 mg, 0.047 mmol) and PMDETA (4.1 mg, 0.023 mmol) in MMA were put into a cuvette, and the mixture was subjected to spectrum scanning at 200–800 nm. Then, the mixture was irradiated with the light for 1 h, and spectrum scanning was performed again.

2.6. Open-to-air chain extension

The macroinitiator was synthesized at a molar ratio of $[MMA]_0$/$[MBPA]_0$/ $[PMDETA]_0$/$[NaI]_0$ = 200/1/1/4 for 6 h ($M_{n,GPC}$ = 6400 g mol^{-1}, M_w/M_n = 1.19) under visible light irradiation. The polymerization and purification processes were carried out according to the procedures described above. In a typical experimental procedure for chain-extension, MMA (468 mg, 4.67 mmol), PMMA (149.0 mg, 0.023 mmol), NaI (14.0 mg, 0.093 mmol), PMDETA (4.1 mg, 0.023 mmol) and 0.5 mL DMAC were placed in a 2 mL ampoule bottle. The solution was not sealed for the polymerization in the presence of air. Polymerization under visible light irradiation (12 W, 4 mW/cm^2) could take place in open air. The reaction time was 5 h.

3. Results and discussion

To verify our hypothesis, we first selected the in situ bromine-iodine transformation as a model system to demonstrate fully oxygen-tolerant RCMP of MMA in the presence of TA, where alkyl iodides were generated in situ by the nucleophilic substitution reaction between MBPA and NaI. Before polymerization, the proton nuclear magnetic resonance (^{1}H NMR) spectra were employed to investigate the bromine-iodine exchange reaction between MBPA and NaI in acetone-d_6. The shift of the signals for H_a (5.70 ppm) adjacent to bromine to those of $H_{a'}$ (5.83 ppm) adjacent to iodide confirmed the gradual formation of alkyl iodide under light irradiation (Fig. 1). Subsequently, photo-induced RCMPs with in

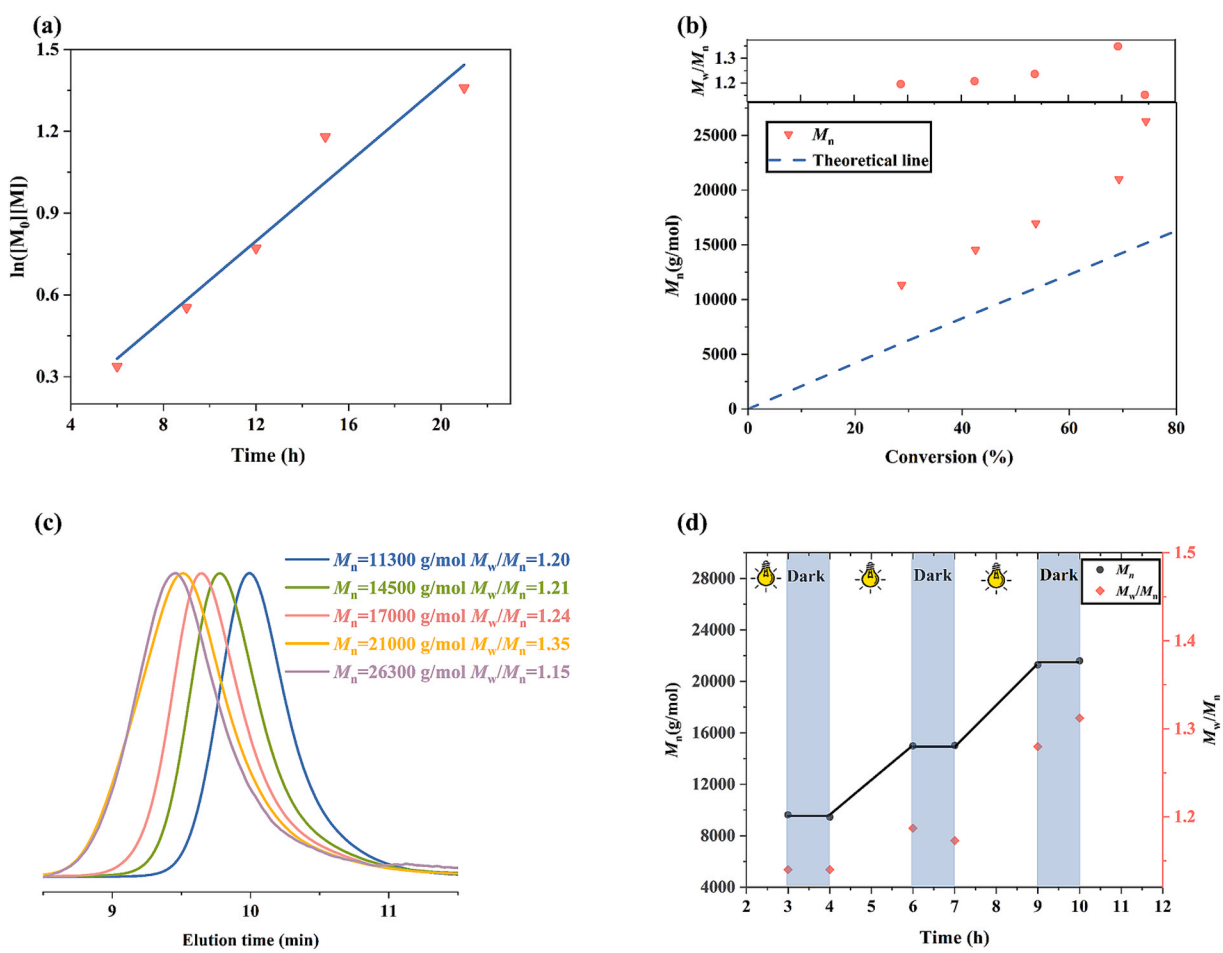

Fig. 2. (a) Ln([M]$_0$/[M]) as a function of time. (b) Molecular weight (M_n) and molecular weight distributions versus conversion. (c) GPC traces evolution with time. (d) Light modulation.

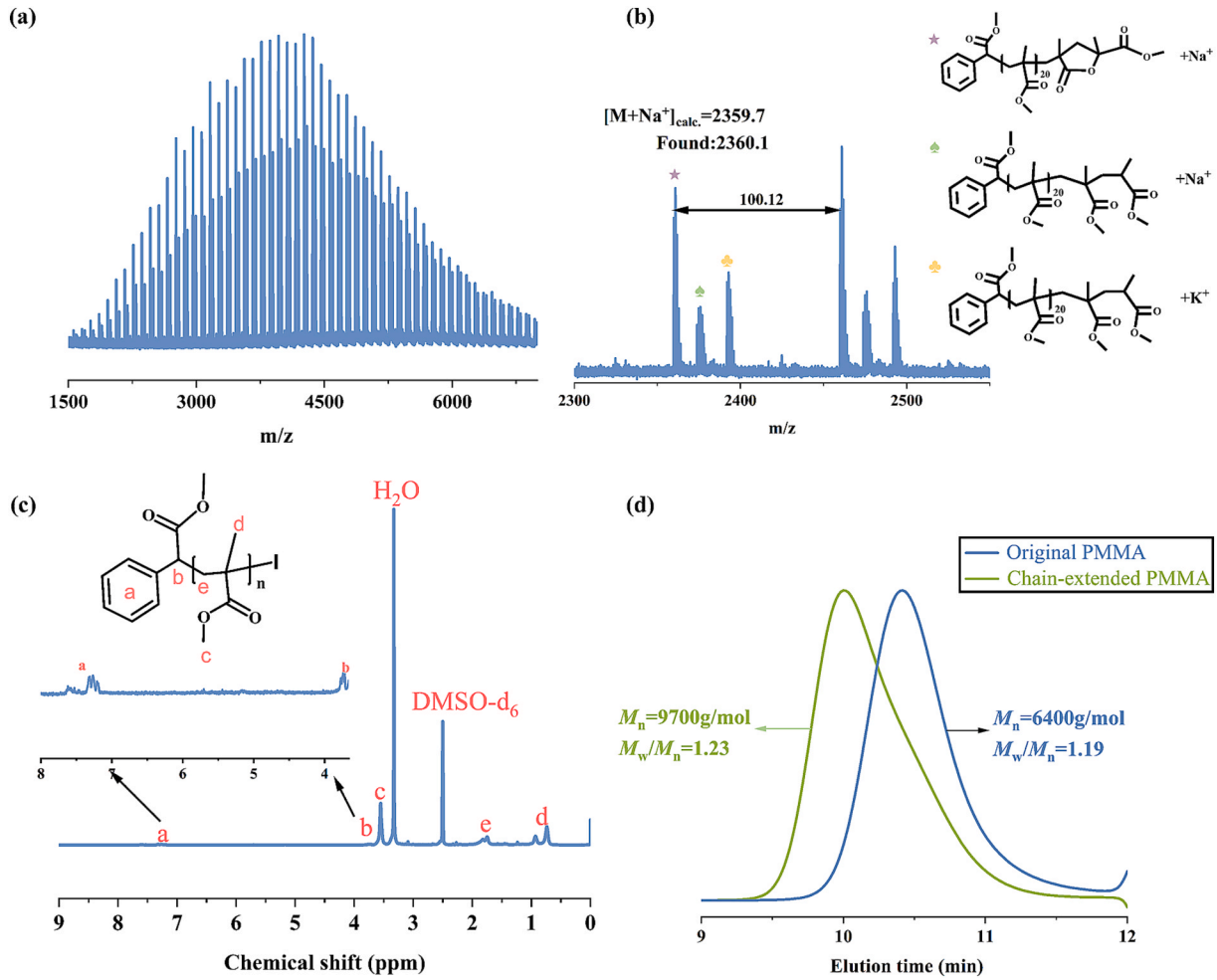

Fig. 3. (a, b) MALDI-TOF-MS analysis of the PMMA (M_n = 4600 g/mol). (c) Typical ^{1}H NMR spectrum of PMMA ($M_{n,\ NMR}$ = 4300 g/mol). (d) GPC curves of PMMA before and after chain extension under visible light irradiation.

situ bromine-iodine transformation were performed under open condition. As expected, photo-induced RCMPs could be successfully conducted under open condition. After 21 h of irradiation, polymethyl methacrylate (PMMA) with a relatively narrow molecular weight distribution (MWD = 1.15) was obtained at high MMA conversion (75 %). Moreover, compared with tetramethylethylenediamine (TMEDA), both PMDETA and tris[2-(dimethylamino) ethyl]amine (Me_6TREN) achieved good control (Table 1, entries 1–3). On the other hand, tetrabutylammonium iodide (TBAI) was used in place of NaI (Table 1, entries 4 and 5), open-to-air RCMPs proceeded effectively, demonstrating that both organic and inorganic iodide salts were capable of regulating polymerization in open air. Besides, it was found that *N,N*-dimethylformamide (DMF), DMSO and *N,N*-dimethylacetamide (DMAC) could be successfully used as solvent for open-to-air RCMPs (Table S1). Among the three solvents evaluated, DMAC provided optimal polymerization control, yielding PMMA with the narrowest molecular weight distribution (M_w/M_n = 1.16) compared to DMF (M_w/M_n = 1.37) and DMSO (M_w/M_n = 1.39). This superior performance likely stems from DMAC's balanced polarity and radical stabilization capability, which effectively maintains the active propagating chains while minimizing side reactions. Thus, DMAC was used in the subsequent discussion of the polymerization. Additionally, it should be noted that a significant deviation between experimental number-average molar masses (M_n) and theoretical values ($M_{n,th}$) was observed in Table 1, which could be attributed to relatively low initiation efficiency in open air.

Encouraged by these promising results, we further explored the kinetic behaviors. The visible light-induced RCMPs with a feed ratio of $[MMA]_0$:$[MBPA]_0$:$[NaI]_0$:$[PMDETA]_0$ = 200:1:4:1 were systematically investigated under open condition. As illustrated in Fig. 2a, a first-order polymerization kinetic was observed, indicating that a reversible equilibrium was still maintained during polymerization process. Noted that polymerization exhibited an induction period (4 h), which could be attributed to the time for formation of initiator and consumption of oxygen. Moreover, the molecular weights of synthesized polymers also increased linearly with monomer conversion, and relatively narrow molecular weight distributions were maintained throughout polymerization (Fig. 2b), suggesting this RCMP exhibited a well-controlled behavior under open condition. However, it should be noted that experimental molecular weights were greatly different from theoretical values. The deviations may be probably caused by the accumulation of I_2 under open condition. In addition, gel permeation chromatography (GPC) curves exhibited a gradual shift to lower retention times with increasing irradiation time (Fig. 2c), consistent with the controlled nature of RDRP. To evaluate temporal control, we performed light ON/OFF cycling experiments. Upon illumination, polymerization smoothly proceeded under aerobic conditions. In contrast, no polymerization occurred in dark, confirming the photoswitchable nature (Fig. 2d).

End-group analysis of PMMA was performed to provide further evidence for the participation of iodinating agent in RCMP. Firstly, matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (MALDI-TOF-MS) revealed several series of peaks with a repeat unit mass of 100.1 Da, corresponding to the theoretical mass of MMA (Fig. 3a and 3b). As expected, the experimental mass closely matched the

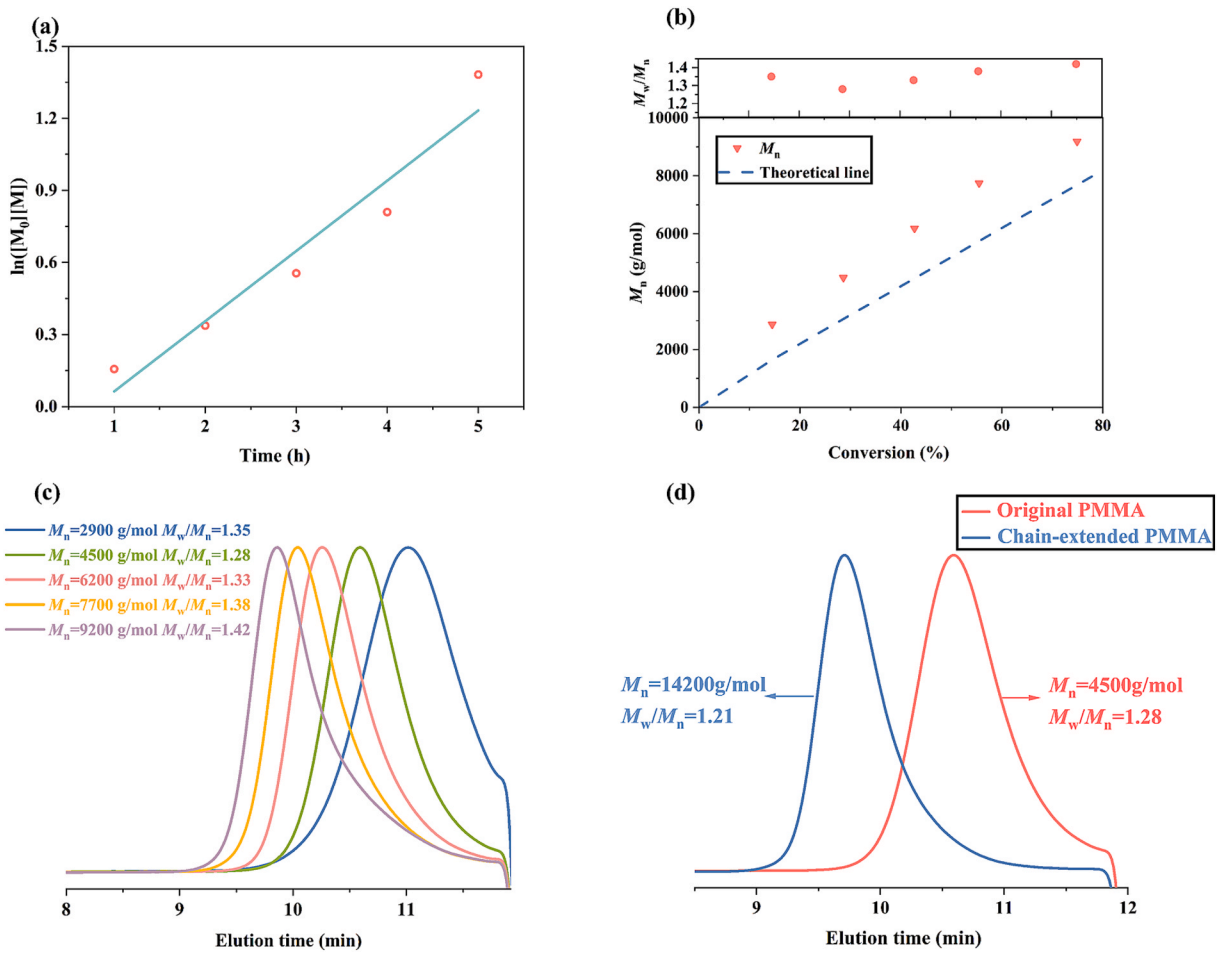

Fig. 4. (a) Ln([M]$_0$/[M]) as a function of time. (b) Molecular weight and molecular weight distributions versus conversion. (c) GPC traces evolution with time. (d) GPC curves of PMMA before and after chain extension under visible light irradiation.

theoretical values. For example, the peak at 2360.1 *m/z* was consistent with theoretical value of 2359.7 g mol^{-1}, assuming that PMMA (DP = 22) end-capped with phenylacetate group and I group at both ends. It is important to point out that the iodinating agent at ω-ends couldnot be directly found because of C-I bond cleavage during the testing process. Three peaks were assigned to PMMA-I and its derivatives, including alkene-terminated PMMA and lactone-terminated PMMA [45–47]. The ^{1}H NMR spectrum revealed the aromatic protons of MBPA at 7.30–8.00 ppm (Fig. 3c), confirming that MBPA was attached to polymer chain ends. Subsequently, open-to-air chain extension experiments were carried out under visible light irradiation. As shown in Fig. 3d, the molecular weight of PMMA increased from 6400 g mol^{-1} to 9700 g mol^{-1}, while molecular weight distributions changed from 1.19 to 1.23, confirming the living characteristics of this open-to-air polymerization.

In order to gain a deeper insight into the polymerization, CP-I was directly employed as both the polymerization initiator and iodinating agent (without MBPA and NaI). The results in Table 1 (entries 6–8) demonstrated that CP-I was successfully photoactivated to regulate open-to-air RCMP in the presence of TA. Among the catalysts tested, PMDETA exhibited the highest efficiency, outperforming TMEDA and Me$_6$TREN. Then, a typical kinetic experiment was conducted using PMDETA as an organocatalyst under open condition (Fig. 4). As shown in Fig. 4a, a linear increase in ln([M]$_0$/[M]) with exposure time was observed, implying that radical concentration remained constant throughout polymerization process. Moreover, an obvious induction period (40 min) was observed, which is characteristic of fully oxygen-tolerant RDRPs. Moreover, the M_n value of obtained polymer increased linearly with monomer conversion, and molecular weight distributions remained relatively narrow (Fig. 4b). In addition, gel permeation chromatography (GPC) curves exhibited a decrease in retention time with irradiation time (Fig. 4c), which further supported the controlled characteristics of this polymerization. Subsequently, open-to-air chain extension experiments were carried out under visible light irradiation. As shown in Fig. 4d, the molecular weight of PMMA increased from 4500 g mol^{-1} to 14200 g mol^{-1}, while molecular weight distributions changed from 1.28 to 1.21. Additionally, MALDI-TOF-MS analysis and ^{1}H NMR spectrum demonstrated good end group fidelity in the current work (Figs. S1 and S2).

In general, iodine radicals tend to recombine with another one to generate an iodine molecule/solvent complex (I_2/solvent). UV–vis spectroscopy of the mixed solution of CP-I and DMAC revealed characteristic absorptions at 366 and 294 nm (Fig. 5a), which were attributed to the I_2/DMAC complex under open-air condition [45]. The observation was further supported by the appearance of a reddish-yellow color after polymerization. Besides, electron spin resonance (ESR) study was employed to detect free radicals by using 5,5-dimethyl-1-pyrroline N-oxide (DMPO) as a spin trapping reagent. As shown in Fig. 5b, the spin adduct formed by trapping the superoxide radical with DMPO was detected, which indicated that oxygen could be reduced to a superoxide radical anion in the present of CP-I and PMDETA, thereby enabling polymerization under aerobic condition [48]. Based on these experimental observations, a possible mechanism for fully oxygen-tolerant RCMP is proposed in Fig. 5c. Under visible light irradiation, tertiary amines can act as electron donors, transferring electron to photoexcited iodides to form iodides radical anions. Moreover, the iodides radical anions can be stabilized by amine radical cations, and take part in the

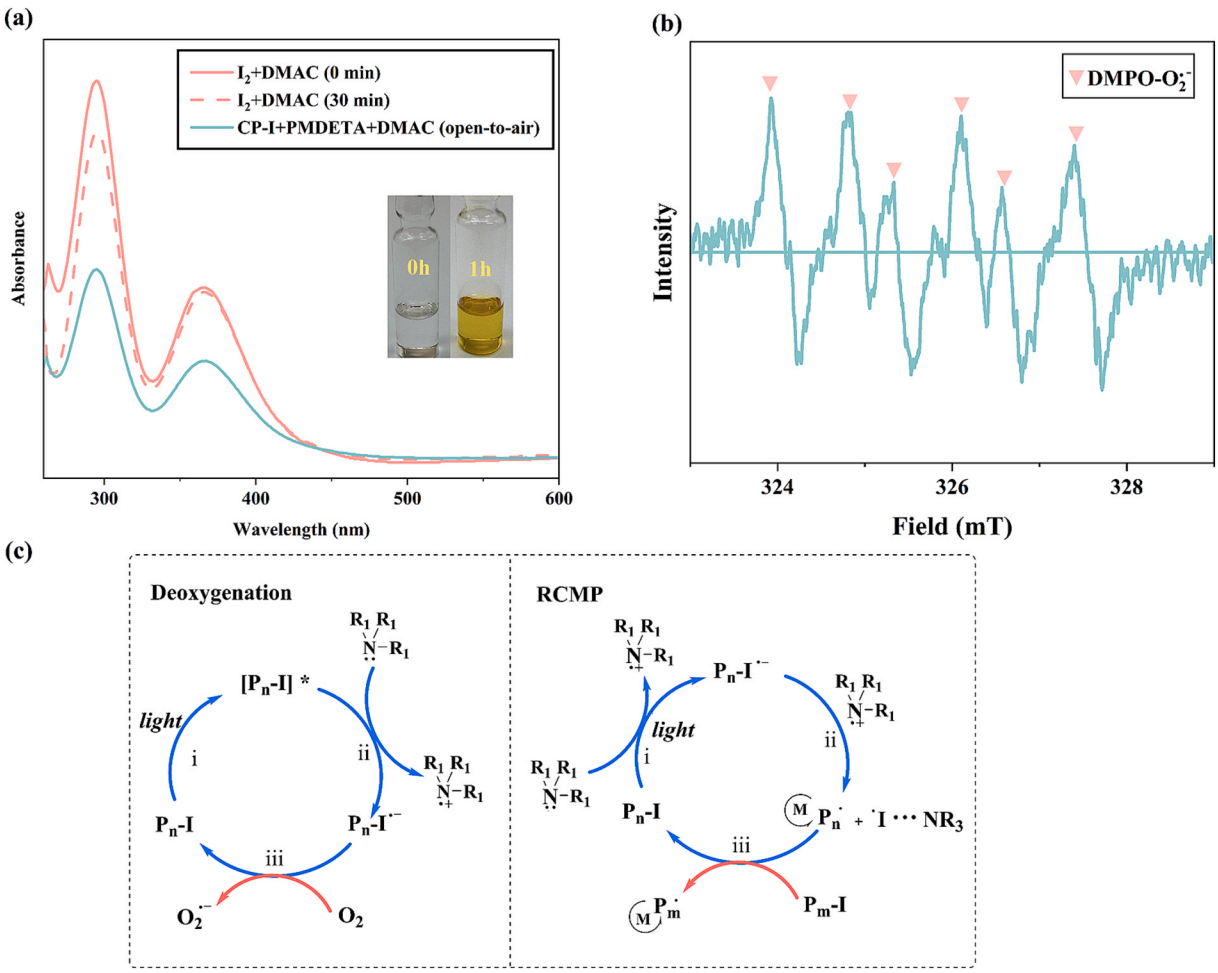

Fig. 5. (a) UV–vis absorption spectra of mixed solutions. (b) ESR spectrum captured by DMPO in DMAC. (c) Proposed mechanism of fully oxygen-tolerant RCMP.

deoxygenation process during polymerization. The iodinating agent mediates electron transfer from the electron donor to convert oxygen (O_2) into an inactive species, successfully overcoming the challenge of using traditional RCMP in an aerobic environment. We deduced that oxygen consumption through the oxidation by the superoxide anion may be the key factor in induction period.

4. Conclusion

In summary, alkyl iodides obtained by bromine-iodine transfer reaction or used directly could be employed to successfully regulate the fully oxygen-tolerant RCMP in the presence of TA. Moreover, MALDI-TOF-MS and ^{1}H NMR data confirmed that alkyl iodides were directly incorporated at the polymer chain ends. Chain-extension experiments further validated the living characteristics of this fully oxygen-tolerant RCMP. We believe that this open-to-air RCMP method offers significant potential for the industrial production of functional polymeric materials.

CRediT authorship contribution statement

Shuang Han: Writing – original draft, Software, Methodology, Investigation. **Yang Guo:** Formal analysis, Software, Investigation. **Fubang Huang:** Methodology, Conceptualization. **Fei Wang:** Software, Methodology. **Weidong Zhang:** Writing – review & editing, Supervision, Project administration, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2025.114124.

Data availability

Data will be made available on request.

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